Name: ________________________

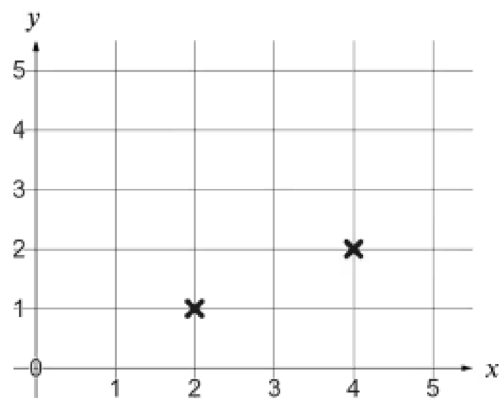
Starter quiz

Representing sequences graphically

1 An arithmetic (linear) sequence has a __________ difference. Tick **1** correct answer

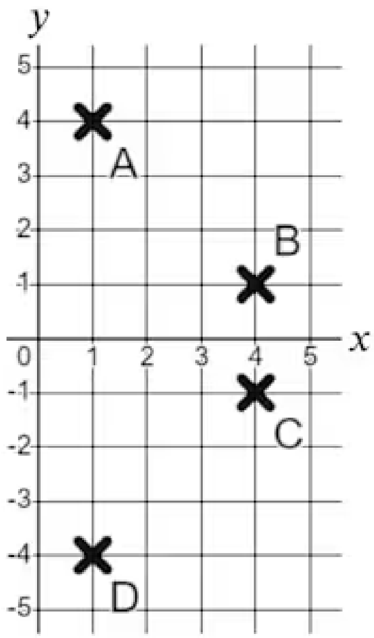
- [ ] constant
- [ ] variable
- [ ] unknown
- [ ] positive

2 Which two pairs of coordinates are shown here? Tick **1** correct answer



- [ ] $(1, 2)$ and $(2, 4)$
- [ ] $(1, 2)$ and $(4, 2)$
- [ ] $(2, 1)$ and $(2, 4)$
- [ ] $(2, 1)$ and $(4, 2)$

3 Match the letters with the correctly plotted coordinates. Write the correct letter in each box



a	A
b	B
c	C
d	D

	$(1, 4)$
	$(1, -4)$
	$(4, 1)$
	$(4, -1)$

4 Which of these are true for this sequence? Tick **3** correct answers

Term number (n)	1	2	3	4	5
Term value (T)	3	5	7	9	11

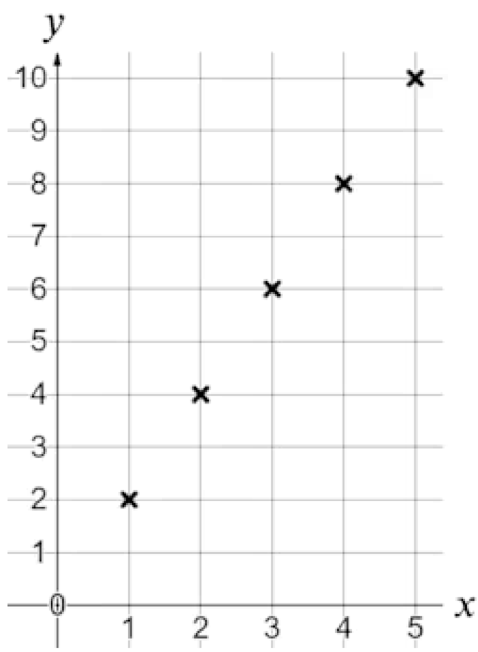
- [ ] When $\mathrm{n} = 1, \mathrm{T} = 3$
- [ ] When $\mathrm{n} = 5, \mathrm{T} = 2$
- [ ] When $\mathrm{n} = 3, \mathrm{T} = 7$
- [ ] When $\mathrm{n} = 5, \mathrm{T} = 5$
- [ ] When $\mathrm{n} = 5, \mathrm{T} = 11$

5 Match the sequences with the common difference. Write the correct letter in each box

a	1, 8, 15, 22, …
b	1, -6, -13, -20, …
c	36, 30, 24, 18, …
d	-39, -33, -27, -21, …
e	0, 3, 7, 12, …
f	-17, -9, -1, 7, …

	+8
	-7
	No common difference
	-6
	+6
	+7

6 What is true of this set of coordinates? Tick **2** correct answers



- ☐ The x coordinate is always double the y coordinate.
- ☐ The y coordinate is always double the x coordinate.
- ☐ The y coordinate is always two more than the x coordinate.
- ☐ As the x coordinate increases by 1, y increases by 2